

# Lecture - 02 VAE, DDPM, Discrete-VAE, D3PMs

① AE - Autoencoder

② VAE - Variational Autoencoder

③ DDPM - Denoising Diffusion Probabilistic Model

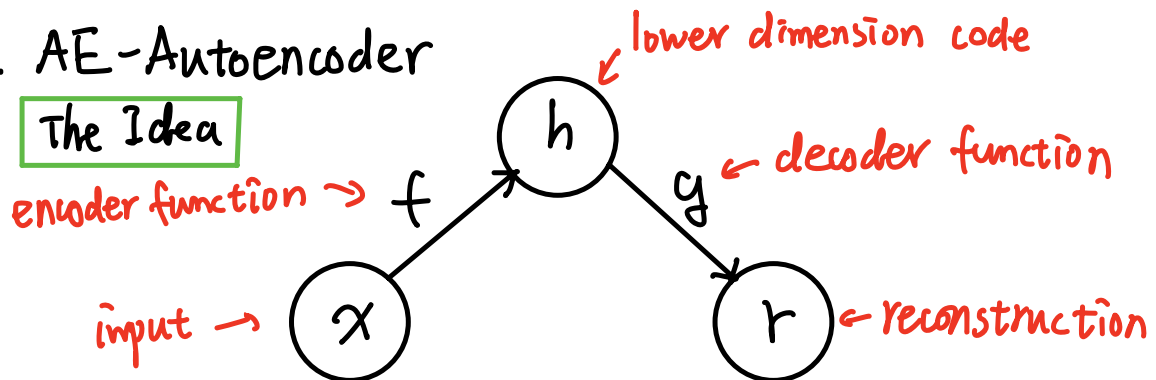
④ Discrete-VAE

⑤ D3PMs

Goal: find  $\phi$  to minimize  $D_{KL}(P_{data}(x) || P_{\phi}(x))$

①. AE - Autoencoder

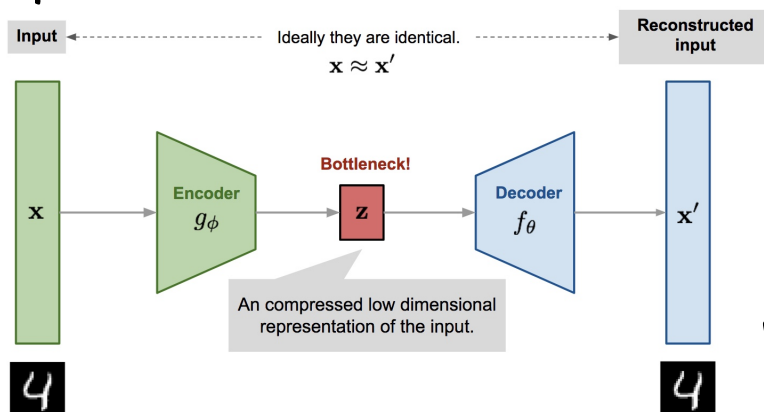
• The Idea



training: attempt to copy its input to its output

$$r = g(f(x)) \Rightarrow \min_{f, g} L(x, g(f(x)))$$

application: dimension reduction, feature learning



• Encoder:

$$x \rightarrow z$$

• Decoder:

$$z \rightarrow x'$$

parameters =  $[\phi, \theta]$

- Ideas of autoencoder originated in 1980s, and later promoted by [Hinton and Salakhutdinov, 2006]
- Application to dimensionality reduction, like PCA
- If  $x$  are images: ( $x \approx f_{\theta}(g_{\phi}(x))$ )

$$L_{AE}(\theta, \phi) = \frac{1}{N} \sum_{i=1}^N (x^{(i)} - f_{\theta}(g_{\phi}(x^{(i)})))^2$$

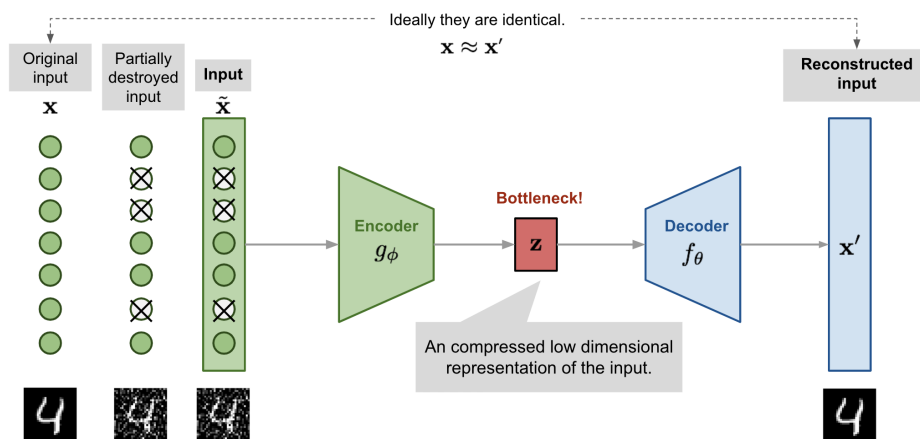
If  $x$  are categorical:

$$L_{AE}(\theta, \phi) = \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^k (-x_k^{(i)} \cdot \log \hat{x}_k^{(i)}),$$

$\hat{x}_k^{(i)}$  is the output of AE. } AE: Easy overfitting

Many variants:

- Denoising autoencoder: Add noise to  $x^{(i)}$



$$L_{DAE}(\theta, \phi) = \frac{1}{N} \sum_{i=1}^N (x^{(i)} - f_{\theta}(g_{\phi}(\tilde{x}^{(i)})))^2$$

$$\tilde{x}^{(i)} \sim \mathcal{M}_0(\tilde{x}^{(i)} | x^{(i)})$$

- Sparse AE
- k-Sparse AE
- Contractive AE

Key limitations:

The latent space of AEs is unstructured: randomly sampling  $z$  produces meaningless outputs.

① AE - Autoencoder

② VAE - Variational Autoencoder

③ DDPM - Denoising Diffusion Probabilistic Model

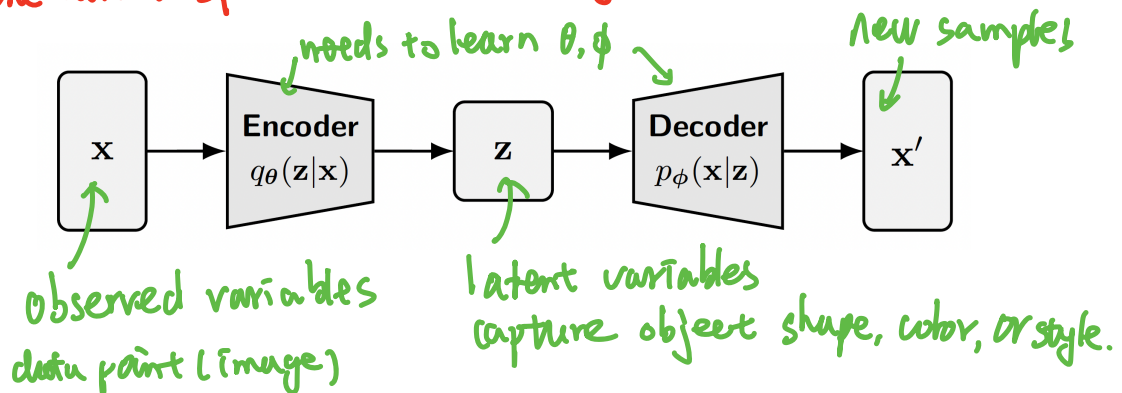
④ Discrete - VAE

⑤ D3PMs

○ New Idea: [Kingma and Welling, 2013]

Auto-encoding Variational Bayes

From  $z \rightarrow p(z)$ : imposing a probabilistic structure on the latent space.  $\rightarrow$  a true generative model.



• generation of  $x'$ :

Step 1:  $z \sim p_{\text{prior}} := \mathcal{N}(0, I)$

Step 2: decoding  $z$  via  $p_{\phi}$ :  $x' \sim p_{\phi}(x|z)$

latent-variable generative model:

$$p_{\phi}(x) = \int p_{\phi}(x|z) \cdot p(z) dz \Rightarrow \text{goal: } \max_{\phi} p_{\phi}(x)$$

intractable, but can use variational method.

- construction of encoder (inference network)

given  $x$ , what latent code  $z$  could have produced it?

$$\overset{\text{posterior}}{\uparrow} P_{\phi}(z|x) = \frac{P_{\phi}(x|z) \cdot p(z)}{P_{\phi}(x)} \leftarrow \text{intractable}$$

Key: "variational" step in VAEs, replacing the intractable posterior with a tractable approximation!

Encoder  $q_{\theta}(z|x)$ : learnable proxy.

$$q_{\theta}(z|x) \approx P_{\phi}(z|x).$$

### Training via ELBO

Theorem.1 Evidence Lower Bound (ELBO)

For any data point  $x$ , the log-likelihood satisfies:

$$\log P_{\phi}(x) \geq \mathcal{L}_{\text{ELBO}}(\theta, \phi; x),$$

$$\mathcal{L}_{\text{ELBO}} = \underbrace{\mathbb{E}_{z \sim q_{\theta}(z|x)} [\log P_{\phi}(x|z)]}_{\text{Reconstruction term}} - \underbrace{D_{\text{KL}}(q_{\theta}(z|x) \parallel p(z))}_{\text{Latent Regularization}}$$

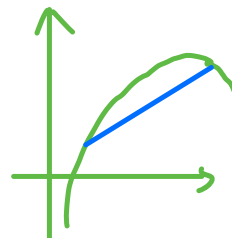
Proof:  $\log P_{\phi}(x) = \log \int P_{\phi}(x, z) dz$

$$= \log \int q_{\theta}(z|x) \cdot \frac{P_{\phi}(x, z)}{q_{\theta}(z|x)} dz$$

$$= \log \mathbb{E}_{z \sim q_{\theta}(z|x)} \left[ \frac{P_{\phi}(x, z)}{q_{\theta}(z|x)} \right]$$

Jensen's Inequality

$$\geq \mathbb{E}_{z \sim q_{\theta}(z|x)} \left[ \log \frac{P_{\phi}(x, z)}{q_{\theta}(z|x)} \right]$$





Note  $p_\phi(x, z) = p_\phi(x|z) \cdot p(z)$

$$\begin{aligned} \log p_\phi(x) &\geq \mathbb{E}_{z \sim q_\theta(z|x)} \left[ \log \frac{p_\phi(x|z) \cdot p(z)}{q_\theta(z|x)} \right] \\ &= \mathbb{E}_{z \sim q_\theta} [\log p_\phi(x|z)] - D_{KL}(q_\theta(z|x) \| p(z)) \end{aligned}$$

Reconstruction: encourages accurate recovery of  $x$  from  $z$

Latent KL: encourages the encoder  $q_\theta(z|x)$  to stay close to  $p_{prior}(z) \rightarrow$  enable meaningful generation.

o Information-theoretic Interpretation

Denote  $\begin{cases} \text{generative joint: } p_\phi(x, z) = p(z) \cdot p_\phi(x|z) \\ \text{inference joint: } q_\theta(x, z) = p_{data}(x) \cdot q_\theta(z|x) \end{cases}$

Recall our goal:  $D_{KL}(p_{data}(x) \| p_\phi(x))$

We have:

$$D_{KL}(p_{data}(x) \| p_\phi(x)) \leq D_{KL}(q_\theta(x, z) \| p_\phi(x, z))$$

↑  
chain rule for KL Divergence

Proof:

$D_{KL}(q_\theta(x, z) \| p_\phi(x, z))$  : Total error bound

$$= \mathbb{E}_{q_\theta(x, z)} \left[ \log \frac{p_{data}(x) \cdot q_\theta(z|x)}{p_\phi(x) \cdot p_\phi(z|x)} \right]$$

$$= \mathbb{E}_{p_{data}(x)} \left[ \log \frac{p_{data}(x)}{p_\phi(x)} + D_{KL}(q_\theta(z|x) \| p_\phi(z|x)) \right]$$

// Note :  $\mathbb{E}_{q_{\theta}(z|x)} = \mathbb{E}_{q_{\theta}(z|x) \cdot p_{\text{data}}(x)}$

$$= \underbrace{D_{KL}(p_{\text{data}} \| p_{\phi})}_{\text{True modeling error} \geq 0} + \underbrace{\mathbb{E}_{p_{\text{data}}(x)} [D_{KL}(q_{\theta}(z|x) \| p_{\phi}(z|x))]}_{\text{Inference error} \geq 0}$$

$$\log p_{\phi}(x) = \log \mathbb{E}_{z \sim q_{\theta}(z|x)} \left[ \frac{p_{\phi}(x, z)}{q_{\theta}(z|x)} \right]$$

$$L_{\text{ELBO}}(\phi, \theta; x) = \mathbb{E}_{z \sim q_{\theta}} [\log p_{\phi}(x|z)] - D_{KL}(q_{\theta}(z|x) \| p(z))$$

$$\log p_{\phi}(x) - L_{\text{ELBO}}(\phi, \theta; x) = D_{KL}(q_{\theta}(z|x) \| p_{\phi}(z|x))$$

gap between log-likelihood  
and the ELBO proxy.

inference error.

maximizing  $L_{\text{ELBO}} \Rightarrow$  reducing inference error.

Proof:

$$\begin{aligned} D_{KL}(q_{\theta}(z|x) \| p_{\phi}(z|x)) &= \int q_{\theta}(z|x) \log \frac{q_{\theta}(z|x)}{p_{\phi}(z|x)} dz \\ &= \int q_{\theta}(z|x) \log \frac{q_{\theta}(z|x) p_{\phi}(x)}{p_{\phi}(z, x)} dz \\ &= \int q_{\theta}(z|x) \left( \log p_{\phi}(x) + \log \frac{q_{\theta}(z|x)}{p_{\phi}(z, x)} \right) dz \\ &\stackrel{*}{=} \log p_{\phi}(x) + \int q_{\theta}(z|x) \log \frac{q_{\theta}(z|x)}{p_{\phi}(z, x)} dz \\ &= \log p_{\phi}(x) + \int q_{\theta}(z|x) \log \frac{q_{\theta}(z|x)}{p_{\phi}(x|z) p_{\phi}(z)} dz \\ &= \log p_{\phi}(x) + \mathbb{E}_{z \sim q_{\theta}(z|x)} \left[ \log \frac{q_{\theta}(z|x)}{p_{\phi}(z)} - \log p_{\phi}(x|z) \right] \\ &= \log p_{\phi}(x) + \underbrace{D_{KL}(q_{\theta}(z|x) \| p_{\phi}(z))}_{\text{inference error}} - \mathbb{E}_{z \sim q_{\theta}(z|x)} [\log p_{\phi}(x|z)], \end{aligned}$$

where  $\stackrel{*}{=}$  is due to  $\int q_{\theta}(z|x) dz = 1$ .

$$-L_{\text{ELBO}}(\phi, \theta; x)$$

$$= \log p_{\phi}(x) - L_{\text{ELBO}}(\phi, \theta; x)$$

## ○ Gaussian VAE

- **Encoder**:  $q_{\theta}(z|x)$  is modeled as a Gaussian

$$q_{\theta}(z|x) := \mathcal{N}(z; \mu_{\theta}(x), \text{diag}(\sigma_{\theta}^2(x))),$$

$$\text{with } \mu_{\theta}: \mathbb{R}^D \rightarrow \mathbb{R}^d, \sigma_{\theta}: \mathbb{R}^D \rightarrow \mathbb{R}_+^d$$

- **Decoder**: modeled as Gaussian

$$p_{\phi}(x|z) := \mathcal{N}(x; \mu_{\phi}(z), \sigma^2 \mathbf{I}),$$

$$\text{with } \mu_{\phi}(z): \mathbb{R}^d \rightarrow \mathbb{R}^D: \mathbb{N}$$

Under this model assumption

$$L_{\text{ELBO}} \left\{ \begin{array}{l} \text{Reconstruction term: } \mathbb{E}_{q_{\theta}(z|x)} [\log p_{\phi}(x|z)] \\ \text{Latent Regularization: } -D_{\text{KL}}(q_{\theta}(z|x) \parallel p_{\text{prior}}(z)) \end{array} \right.$$

- Reconstruction:

$$\mathbb{E}_{q_{\theta}(z|x)} [\log p_{\phi}(x|z)] = -\frac{1}{2\sigma^2} \mathbb{E}_{q_{\theta}(z|x)} [\|x - \mu_{\phi}(z)\|^2] + C$$

↓ ELBO objective

$$\min_{\theta, \phi} \mathbb{E}_{q_{\theta}(z|x)} \left[ \frac{1}{2\sigma^2} \|x - \mu_{\phi}(z)\|^2 \right] + D_{\text{KL}}(q_{\theta}(z|x) \parallel p_{\text{prior}}(z))$$

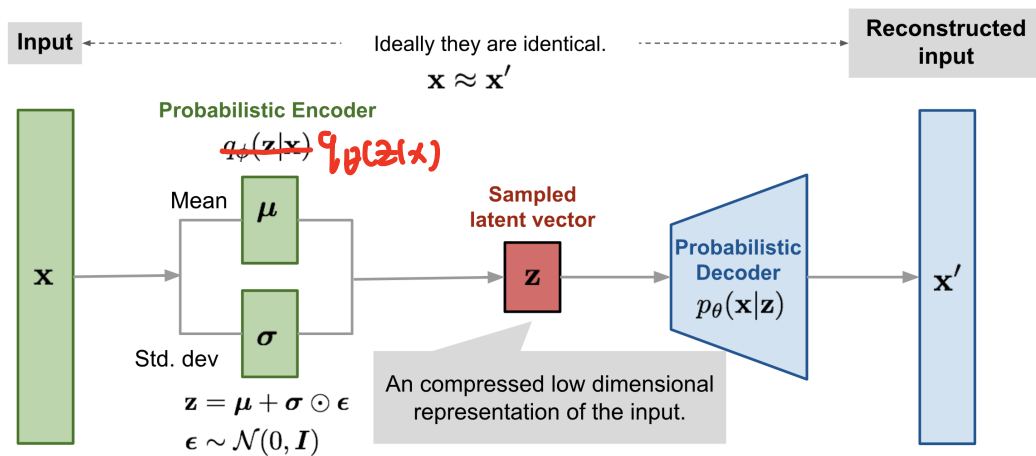
**Proof:** For D-dimensional Gaussian with mean  $\mu$  and

covariance  $\sigma^2 \mathbf{I}$ :

$$p(x) = \frac{1}{(2\pi\sigma^2)^{D/2}} \cdot \exp\left(-\frac{1}{2\sigma^2} \|x - \mu\|^2\right)$$

$$\log p(x) = -\frac{1}{2\sigma^2} \|x - \mu\|^2 - \frac{D}{2} \log(2\pi\sigma^2).$$

Replacing  $\mu$  with  $\mu_{\phi}(z)$  and  $C = -\frac{D}{2} \log(2\pi\sigma^2)$  ■



•  $q_{\theta}(z|x)$  For sample  $x^{(i)}$ :

$$z \sim q_{\theta}(z|x^{(i)}) = \mathcal{N}(z; \mu_i, \sigma_i^2 \cdot I)$$

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

### Drawbacks of Standard VAE

VAEs often produces blurry outputs

Consider:  $\begin{cases} \text{fixed Gaussian encoder } q_{\text{enc}}(z|x) \\ \text{decoder } p_{\text{dec}}(x|z) = \mathcal{N}(x; \mu(z), \sigma^2 I) \end{cases}$

$$\Rightarrow \text{ELBO: } \underset{\mu}{\text{arg max}} \mathbb{E}_{p_{\text{data}}(x) \cdot q_{\text{enc}}(z|x)} [\|x - \mu(z)\|^2]$$

$$\Rightarrow \mu^*(z) = \mathbb{E}_{q_{\text{enc}}(x|z)} [x] \quad \text{with}$$

$$q_{\text{enc}}(x|z) = \frac{q_{\text{enc}}(z|x) \cdot p_{\text{data}}(x)}{p_{\text{prior}}(z)}$$

$$\Rightarrow \mu^*(z) = \frac{\mathbb{E}_{p_{\text{data}}(x)} [q_{\text{enc}}(z|x) \cdot x]}{\mathbb{E}_{p_{\text{data}}(x)} [q_{\text{enc}}(z|x)]}$$

↑ missing normal?

If two inputs  $x \neq x'$ , such that  $q_{\text{enc}}(z|x)$

$q_{\text{enc}}(z|x')$  overlap

$\Rightarrow$  share  $z \rightarrow$  blurry

**Proof:**  $\mathbb{E}_{x,z} [f(x,z)] = \mathbb{E}_z [\mathbb{E}_{x|z} [f(x,z)]]$ ,

with  $f(x,z) = \|x - \mu(z)\|^2$

$$\mathbb{E}_{p(x)} \mathbb{E}_{q(z|x)} [\cdot] = \mathbb{E}_{q(z)} \mathbb{E}_{q(x|z)} [\cdot]$$

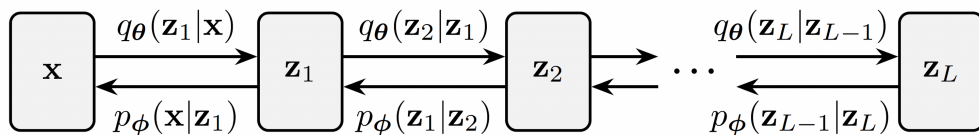
$$L(\mu) = \mathbb{E}_{q(z)} [\mathbb{E}_{q(x|z)} [\|x - \mu(z)\|^2]]$$

$$\arg \min_{\mu} L(\mu) = \arg \min_{\mu} \mathbb{E}_{q(x|z)} [\|x - \mu(z)\|^2]$$

Each subproblem:

$$\min_{\mu(z)} \mathbb{E}_{q(x|z)} [\|x - \mu(z)\|^2] \Leftrightarrow \mu^*(z) = \mathbb{E}_{q(x|z)} [x]$$

## From VAE to Hierarchical VAEs



HVAEs introduce multiple layers of latent code  $z$

$$p_{\phi}(x, z_{1:L}) = p_{\phi}(x|z_1) \cdot \prod_{i=1}^L p_{\phi}(z_{i+1}|z_i) \cdot p(z_L)$$

$$p_{\text{HVAE}}(x) := \int p_{\phi}(x, z_{1:L}) dz_{1:L}$$

$$q_{\theta}(z_{1:L}|x) = q_{\theta}(z_1|x) \cdot \prod_{i=1}^L q_{\theta}(z_{i+1}|z_i)$$

HVAE's ELBO:

$$\begin{aligned}
\log p_{\text{HVAE}}(\mathbf{x}) &= \log \int p_{\phi}(\mathbf{x}, \mathbf{z}_{1:L}) d\mathbf{z}_{1:L} \\
&= \log \int \frac{p_{\phi}(\mathbf{x}, \mathbf{z}_{1:L})}{q_{\theta}(\mathbf{z}_{1:L}|\mathbf{x})} q_{\theta}(\mathbf{z}_{1:L}|\mathbf{x}) d\mathbf{z}_{1:L} \\
&= \log \mathbb{E}_{q_{\theta}(\mathbf{z}_{1:L}|\mathbf{x})} \left[ \frac{p_{\phi}(\mathbf{x}, \mathbf{z}_{1:L})}{q_{\theta}(\mathbf{z}_{1:L}|\mathbf{x})} \right] \\
&\geq \mathbb{E}_{q_{\theta}(\mathbf{z}_{1:L}|\mathbf{x})} \left[ \log \frac{p_{\phi}(\mathbf{x}, \mathbf{z}_{1:L})}{q_{\theta}(\mathbf{z}_{1:L}|\mathbf{x})} \right] \\
&=: \mathcal{L}_{\text{ELBO}}(\phi).
\end{aligned}$$

Substituting the factorized forms yields:

$$\mathcal{L}_{\text{ELBO}} = \mathbb{E}_{q_{\theta}(\mathbf{z}_{1:L}|\mathbf{x})} \left[ \log \frac{p(\mathbf{z}_L) \prod_{i=2}^L p_{\phi}(\mathbf{z}_{i-1}|\mathbf{z}_i) p_{\phi}(\mathbf{x}|\mathbf{z}_1)}{q_{\theta}(\mathbf{z}_1|\mathbf{x}) \prod_{i=2}^L q_{\theta}(\mathbf{z}_i|\mathbf{z}_{i-1})} \right].$$

### Observation 2 (Key observations from HVAE)

*Stacking layers allows the model to generate data progressively, starting with coarse details and adding finer ones at each step. This process makes it far easier to capture the complex structure of high-dimensional data.*



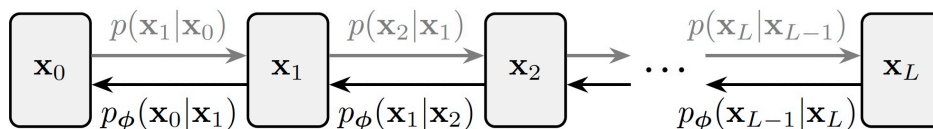
Figure 1: 256×256-pixel samples generated by NVAE, trained on CelebA HQ [28].

• Deeper Networks in a flat VAE are not enough.

①. variational family is limited:

- Greater network depth improves  $\mu_{\theta}$  and  $\sigma_{\theta}$  but does not expand the family
- if the decoder is too expensive, the model may suffer from posterior collapse

- ① AE – Autoencoder
- ② VAE – Variational Autoencoder
- ③ DDPM – Denoising Diffusion Probabilistic Model
- ④ Discrete – VAE
- ⑤ D3PMs

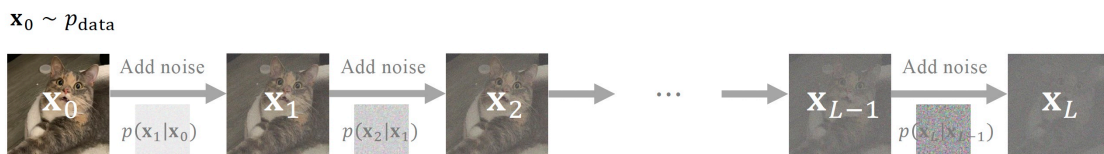


DDPMs { ①. Forward Pass (Fixed Encoder)  
 ②. Reverse Denoising Process (Learnable Decoder)

• DDPMs achieve remarkable stability and expressive power.

### Forward Process (Fixed Encoder)

$$x_0 \sim p_{\text{data}} \longrightarrow x_L \sim p_{\text{prior}} := \mathcal{N}(0, I)$$



Each step is governed by a fixed Gaussian transitional kernel

$$p(x_i | x_{i-1}) := \mathcal{N}(x_i | \sqrt{1 - \beta_i^2} x_{i-1}, \beta_i^2 I),$$

$i = 1, 2, \dots, L$  with  $x_0 \sim p_{\text{data}}$ .

$\{\beta_i\}_{i=1}^L$  : pre-determined or learned,  $\beta_i \uparrow$  and  $\beta_i \in (0, 1)$

Define  $\alpha_i := \sqrt{1 - \beta_i^2}$ . Based on the definition of  $p(x_i | x_{i-1})$ ,

$$x_i = \alpha_i \cdot x_{i-1} + \beta_i \cdot \epsilon_i, \quad \epsilon_i \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, I)$$

↑  
controlled noise

[ This is from the fact of sampling from a Gaussian  
If  $X \sim \mathcal{N}(\mu, \sigma^2 I)$ , then the sample can be generated  
as:  $X = \mu + \sigma \cdot \epsilon$ ,  $\epsilon \sim \mathcal{N}(0, I)$ . ]

Applying the transition kernels recursively:

$$x_1 = \alpha_1 x_0 + \beta_1 \epsilon_1$$

$$x_2 = \alpha_2 x_1 + \beta_2 \epsilon_2$$

$$= \alpha_2 (\alpha_1 x_0 + \beta_1 \epsilon_1) + \beta_2 \epsilon_2$$

$$= \alpha_2 \alpha_1 x_0 + \alpha_2 \beta_1 \epsilon_1 + \beta_2 \epsilon_2$$

$$x_3 = \alpha_3 x_2 + \beta_3 \epsilon_3$$

$$\vdots$$

$$= \alpha_3 \alpha_2 \alpha_1 x_0 + \alpha_3 \alpha_2 \beta_1 \epsilon_1 + \alpha_3 \beta_2 \epsilon_2 + \beta_3 \epsilon_3$$

$$\Rightarrow x_i = \left( \prod_{j=1}^i \alpha_j \right) \cdot x_0 + \sum_{k=1}^i \left( \beta_k \cdot \prod_{j=k+1}^i \alpha_j \right) \cdot \epsilon_k$$

Note, they are i.i.d.

Define  $\bar{\alpha}_i = \prod_{j=1}^i \alpha_j$

//  $\sum_{k=1}^i \beta_k \cdot \frac{\bar{\alpha}_i}{\bar{\alpha}_k}$

To show  $\text{Var} \left[ \sum_{k=1}^i \beta_k \cdot \frac{\bar{\alpha}_i}{\bar{\alpha}_k} \epsilon_k \right] = 1 - \bar{\alpha}_i^2$  telescoping

Note  $1 - \bar{\alpha}_i^2 = 1 - \prod_{j=1}^i \alpha_j^2$  variance and  $\begin{cases} 1 - \alpha_i^2 = \beta_i^2 \\ 1 - \alpha_i^2 \alpha_{i-1}^2 = (1 - \alpha_i^2) + (1 - \alpha_{i-1}^2) \alpha_i^2 \\ 1 - \alpha_i^2 \alpha_{i-1}^2 \alpha_{i-2}^2 = (1 - \alpha_i^2) + (1 - \alpha_{i-1}^2) \alpha_i^2 + (1 - \alpha_{i-2}^2) \alpha_i^2 \alpha_{i-1}^2 \end{cases}$  telescoping

↓

$$\sum_{k=1}^i \left( \beta_k \cdot \frac{\bar{\alpha}_i}{\bar{\alpha}_k} \right)^2 = \sum_{k=1}^i (1 - \alpha_k^2) \cdot \frac{\bar{\alpha}_i^2}{\bar{\alpha}_k^2}$$

So, the mean is  $\bar{\alpha}_i \cdot x_0$  and variance is  $(1 - \bar{\alpha}_i^2) \cdot I$



Perturbation Kernel and Prior Distribution:

$$p_i(x_i | x_0) = \mathcal{N}(x_i; \bar{\alpha}_i \cdot x_0, (1 - \bar{\alpha}_i^2) \cdot \mathbf{I}),$$

$$\bar{\alpha}_i = \prod_{k=1}^i \alpha_k = \prod_{k=1}^i \sqrt{1 - \beta_k^2} = \prod_{k=1}^i \alpha_k$$

Sampling:  $x_i = \bar{\alpha}_i \cdot x_0 + \sqrt{1 - \bar{\alpha}_i^2} \cdot \xi, \xi \sim \mathcal{N}(0, \mathbf{I})$

Since  $\{\beta_i\}_{i=1}^L \nearrow$  and  $\beta_i \in (0, 1), \beta_L \rightarrow 1$ .

$$\Rightarrow \sqrt{1 - \beta_L^2} \rightarrow 0, \bar{\alpha}_i \rightarrow 0 \Rightarrow \sqrt{1 - \bar{\alpha}_i^2} \rightarrow 1.$$

$$p_L(x_L | x_0) \rightarrow \mathcal{N}(0, \mathbf{I}) \text{ as } L \rightarrow \infty,$$

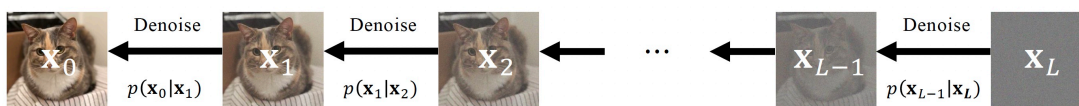
which motivates the choice of the Pprior as:

$$p_{\text{prior}} := \mathcal{N}(0, \mathbf{I}) \Leftrightarrow \text{No reliance on data } x_0.$$

## Reverse Denoising Process (Learnable Decoder)

$x_L \sim p_{\text{prior}} \xrightarrow{\text{denoise}} x_0$ : meaningful data point

$x_0 \sim p_{\text{data}}$



Key Q: Can we effectively approximate these reverse transition kernels  $p(x_{i-1}|x_i)$  especially when  $x_i \sim p_i(x_i)$  is complex?

To approximate  $p(x_{i-1}|x_i)$ , we use  $p_\phi(x_{i-1}|x_i)$  and minimize

$$(*) \quad \min_{\phi} \mathbb{E}_{p_i(x_i)} [\mathcal{D}_{KL}(p(x_{i-1}|x_i) || p_\phi(x_{i-1}|x_i))]$$

Note  $p(x_{i-1}|x_i) = p(x_i|x_{i-1}) \cdot \frac{p_{i-1}(x_{i-1})}{p_i(x_i)}$  ← intractable

as  $p_i(x_i) = \int p_i(x_i | x_0) \cdot \underbrace{p_{\text{data}}(x_0)}_{\text{no closed-form}} dx_0$

Key Idea: we condition the reverse transition on a clean data sample  $x$ .  $\rightarrow$  make the kernel tractable.

$\Rightarrow p(x_{i-1} | x_i, x) = \underbrace{p(x_i | x_{i-1}, x)}_{\text{it has closed form.}} \cdot \frac{p(x_{i-1} | x)}{p(x_i | x)}$

(Markov property of  $p(x_i | x_{i-1}, x) = p(x_i | x_{i-1})$  from the forward process assumption)

(\*)  $\Rightarrow$  [Marginal KL Minimization  $\Leftrightarrow$  Conditional KL Min.]

#### Theorem 2.2.1: Equivalence Between Marginal and Conditional KL Minimization

The following equality holds:

$$\begin{aligned} & \mathbb{E}_{p_i(\mathbf{x}_i)} [\mathcal{D}_{\text{KL}}(p(\mathbf{x}_{i-1} | \mathbf{x}_i) \| p_\phi(\mathbf{x}_{i-1} | \mathbf{x}_i))] \\ &= \mathbb{E}_{p_{\text{data}}(\mathbf{x})} \mathbb{E}_{p(\mathbf{x}_i | \mathbf{x})} [\mathcal{D}_{\text{KL}}(p(\mathbf{x}_{i-1} | \mathbf{x}_i, \mathbf{x}) \| p_\phi(\mathbf{x}_{i-1} | \mathbf{x}_i))] + C, \end{aligned} \quad (2.2.3)$$

where  $C$  is a constant independent of  $\phi$ . Moreover, the minimizer of Equation (2.2.3) satisfies

$$p^*(\mathbf{x}_{i-1} | \mathbf{x}_i) = \mathbb{E}_{p(\mathbf{x} | \mathbf{x}_i)} [p(\mathbf{x}_{i-1} | \mathbf{x}_i, \mathbf{x})] = p(\mathbf{x}_{i-1} | \mathbf{x}_i), \quad \mathbf{x}_i \sim p_i.$$

#### Proof. Derivation of Equation (2.2.3).

We start by expanding the right-hand side expectation:

$$\begin{aligned} & \mathbb{E}_{p(\mathbf{x}_0, \mathbf{x}_i)} [\mathcal{D}_{\text{KL}}(p(\mathbf{x}_{i-1} | \mathbf{x}_i, \mathbf{x}_0) \| p_\phi(\mathbf{x}_{i-1} | \mathbf{x}_i))] \\ &= \int \int p(\mathbf{x}_0, \mathbf{x}_i) \mathcal{D}_{\text{KL}}(p(\mathbf{x}_{i-1} | \mathbf{x}_i, \mathbf{x}_0) \| p_\phi(\mathbf{x}_{i-1} | \mathbf{x}_i)) d\mathbf{x}_0 d\mathbf{x}_i. \end{aligned}$$

By the definition of KL divergence,

$$\mathcal{D}_{\text{KL}}(p(\mathbf{x}_{i-1} | \mathbf{x}_i, \mathbf{x}_0) \| p_\phi(\mathbf{x}_{i-1} | \mathbf{x}_i)) = \int p(\mathbf{x}_{i-1} | \mathbf{x}_i, \mathbf{x}_0) \log \frac{p(\mathbf{x}_{i-1} | \mathbf{x}_i, \mathbf{x}_0)}{p_\phi(\mathbf{x}_{i-1} | \mathbf{x}_i)} d\mathbf{x}_{i-1}.$$

Substituting this into the expectation, we have

$$\int \int \int p(\mathbf{x}_0, \mathbf{x}_i) p(\mathbf{x}_{i-1} | \mathbf{x}_i, \mathbf{x}_0) \log \frac{p(\mathbf{x}_{i-1} | \mathbf{x}_i, \mathbf{x}_0)}{p_\phi(\mathbf{x}_{i-1} | \mathbf{x}_i)} d\mathbf{x}_{i-1} d\mathbf{x}_0 d\mathbf{x}_i.$$

Using the chain rule of probability,

$$p(\mathbf{x}_0, \mathbf{x}_i) = p(\mathbf{x}_i) p(\mathbf{x}_0 | \mathbf{x}_i),$$

we rewrite the integral as

$$\int p(\mathbf{x}_i) \int p(\mathbf{x}_0 | \mathbf{x}_i) \int p(\mathbf{x}_{i-1} | \mathbf{x}_i, \mathbf{x}_0) \log \frac{p(\mathbf{x}_{i-1} | \mathbf{x}_i, \mathbf{x}_0)}{p_\phi(\mathbf{x}_{i-1} | \mathbf{x}_i)} d\mathbf{x}_{i-1} d\mathbf{x}_0 d\mathbf{x}_i.$$

This allows us to express the expectation in nested form:

$$\mathbb{E}_{p(\mathbf{x}_i)} \left[ \mathbb{E}_{p(\mathbf{x}_0|\mathbf{x}_i)} \left[ \mathbb{E}_{p(\mathbf{x}_{i-1}|\mathbf{x}_i, \mathbf{x}_0)} \left[ \log \frac{p(\mathbf{x}_{i-1}|\mathbf{x}_i, \mathbf{x}_0)}{p_\phi(\mathbf{x}_{i-1}|\mathbf{x}_i)} \right] \right] \right].$$

Next, we apply the decomposition of the logarithm:

$$\log \frac{p(\mathbf{x}_{i-1}|\mathbf{x}_i, \mathbf{x}_0)}{p_\phi(\mathbf{x}_{i-1}|\mathbf{x}_i)} = \log \frac{p(\mathbf{x}_{i-1}|\mathbf{x}_i, \mathbf{x}_0)}{\underbrace{p(\mathbf{x}_{i-1}|\mathbf{x}_i)}} + \log \frac{\underbrace{p(\mathbf{x}_{i-1}|\mathbf{x}_i)}}{p_\phi(\mathbf{x}_{i-1}|\mathbf{x}_i)}.$$

Substituting this back into the expectation gives two terms:

$$\begin{aligned} & \mathbb{E}_{p(\mathbf{x}_i)} \left[ \mathbb{E}_{p(\mathbf{x}_0|\mathbf{x}_i)} \left[ \mathbb{E}_{p(\mathbf{x}_{i-1}|\mathbf{x}_i, \mathbf{x}_0)} \left[ \log \frac{p(\mathbf{x}_{i-1}|\mathbf{x}_i, \mathbf{x}_0)}{p(\mathbf{x}_{i-1}|\mathbf{x}_i)} \right] \right] \right] \\ & + \mathbb{E}_{p(\mathbf{x}_i)} \left[ \mathbb{E}_{p(\mathbf{x}_0|\mathbf{x}_i)} \left[ \mathbb{E}_{p(\mathbf{x}_{i-1}|\mathbf{x}_i, \mathbf{x}_0)} \left[ \log \frac{p(\mathbf{x}_{i-1}|\mathbf{x}_i)}{p_\phi(\mathbf{x}_{i-1}|\mathbf{x}_i)} \right] \right] \right]. \end{aligned}$$

Since the second logarithmic term does not depend on  $\mathbf{x}_0$ , by the law of total probability

$$\mathbb{E}_{p(\mathbf{x}_0|\mathbf{x}_i)} \left[ \mathbb{E}_{p(\mathbf{x}_{i-1}|\mathbf{x}_i, \mathbf{x}_0)} \left[ \log \frac{p(\mathbf{x}_{i-1}|\mathbf{x}_i)}{p_\phi(\mathbf{x}_{i-1}|\mathbf{x}_i)} \right] \right] = \mathbb{E}_{p(\mathbf{x}_{i-1}|\mathbf{x}_i)} \left[ \log \frac{p(\mathbf{x}_{i-1}|\mathbf{x}_i)}{p_\phi(\mathbf{x}_{i-1}|\mathbf{x}_i)} \right].$$

Similarly, the first term is the KL divergence

$$\mathbb{E}_{p(\mathbf{x}_0|\mathbf{x}_i)} [\mathcal{D}_{\text{KL}}(p(\mathbf{x}_{i-1}|\mathbf{x}_i, \mathbf{x}_0) \| p(\mathbf{x}_{i-1}|\mathbf{x}_i))].$$

Putting it all together, we obtain the decomposition:

$$\begin{aligned} & \mathbb{E}_{p(\mathbf{x}_0, \mathbf{x}_i)} [\mathcal{D}_{\text{KL}}(p(\mathbf{x}_{i-1}|\mathbf{x}_i, \mathbf{x}_0) \| p_\phi(\mathbf{x}_{i-1}|\mathbf{x}_i))] \\ & = \mathbb{E}_{p(\mathbf{x}_i)} \left[ \mathbb{E}_{p(\mathbf{x}_0|\mathbf{x}_i)} [\mathcal{D}_{\text{KL}}(p(\mathbf{x}_{i-1}|\mathbf{x}_i, \mathbf{x}_0) \| p(\mathbf{x}_{i-1}|\mathbf{x}_i))] \right] \\ & \quad + \mathbb{E}_{p(\mathbf{x}_i)} [\mathcal{D}_{\text{KL}}(p(\mathbf{x}_{i-1}|\mathbf{x}_i) \| p_\phi(\mathbf{x}_{i-1}|\mathbf{x}_i))]. \end{aligned}$$

**Proof of Optimality.** To prove:

$$p_\phi^*(\mathbf{x}_{i-1}|\mathbf{x}_i) = p(\mathbf{x}_{i-1}|\mathbf{x}_i) = \mathbb{E}_{p(\mathbf{x}|\mathbf{x}_i)} [p(\mathbf{x}_{i-1}|\mathbf{x}_i, \mathbf{x})], \quad \mathbf{x}_i \sim p_i.$$

The first identity follows from the fact that the KL divergence  $\mathcal{D}_{\text{KL}}(p \| p_\phi)$  is minimized when  $p^* = p$ , assuming the parameterization is sufficiently expressive. The second identity follows directly from the law of total probability.

■

Conditioning to obtain a tractable objective, forms the foundation of DDPMs.

**Lemma 2.2.2: Reverse Conditional Transition Kernel**

$p(\mathbf{x}_{i-1}|\mathbf{x}_i, \mathbf{x})$  is Gaussian with the closed-form expression:

$$p(\mathbf{x}_{i-1}|\mathbf{x}_i, \mathbf{x}) = \mathcal{N}\left(\mathbf{x}_{i-1}; \boldsymbol{\mu}(\mathbf{x}_i, \mathbf{x}, i), \sigma^2(i)\mathbf{I}\right),$$

where

$$\boldsymbol{\mu}(\mathbf{x}_i, \mathbf{x}, i) := \frac{\bar{\alpha}_{i-1}\beta_i^2}{1 - \bar{\alpha}_i^2}\mathbf{x} + \frac{(1 - \bar{\alpha}_{i-1}^2)\alpha_i}{1 - \bar{\alpha}_i^2}\mathbf{x}_i, \quad \sigma^2(i) := \frac{1 - \bar{\alpha}_{i-1}^2}{1 - \bar{\alpha}_i^2}\beta_i^2.$$

We will be back  
to this lemma in  
Lemma 4.4.2

## Modeling of reverse transition kernel $p_\phi(x_{i-1}|x_i)$

DDPM, assume  $p_\phi(x_{i-1}|x_i)$  is Gaussian:

$$p_\phi(x_{i-1}|x_i) := \mathcal{N}(x_{i-1}; \mu_\phi(x_i, i), \sigma^2(i) \cdot \mathbf{I})$$

- $\mu_\phi(\cdot, i) : \mathbb{R}^D \rightarrow \mathbb{R}^D$  learnable mean function
- $\sigma^2(i) := \frac{1 - \bar{\alpha}_i^2}{1 - \bar{\alpha}_{i-1}^2} \cdot \beta_i^2$ . To match all layers, our loss is:

$$\mathcal{L}_{\text{diff}}(x_0; \phi) := \sum_{i=1}^L \mathbb{E}_{p(x_i|x_0)} [\mathcal{D}_{\text{KL}}(p_i, p_{i,\phi})],$$

$$\mathcal{D}_{\text{KL}}(p_i, p_{i,\phi}) = \mathcal{D}_{\text{KL}}(p(x_{i-1}|x_i, x_0) \parallel p_\phi(x_{i-1}|x_i)).$$

Given  $p_1 \sim \mathcal{N}(\mu_1, \Sigma_1)$ ,  $p_2 \sim \mathcal{N}(\mu_2, \Sigma_2)$ , the KL divergence from  $p_2 \rightarrow p_1$  is:

$$\mathcal{D}_{\text{KL}}(p_1, p_2) = \frac{1}{2} \left\{ \text{tr}(\Sigma_2^{-1} \Sigma_1) + (\mu_2 - \mu_1)^T \cdot \Sigma_2^{-1} \cdot (\mu_2 - \mu_1) - D + \ln \frac{|\Sigma_2|}{|\Sigma_1|} \right\}$$

replace  $\mu_1$  with  $\mu$ ,  $\mu_2$  with  $\mu_\phi$ ,  $\Sigma_1 = \Sigma_2 = \sigma^2(i) \cdot \mathbf{I}$ .

$$\mathcal{D}_{\text{KL}}(p_i, p_{i,\phi}) = \frac{1}{2} \left\{ D + \frac{1}{\sigma^2(i)^2} \cdot \|\mu - \mu_\phi\|^2 - D + \ln 1 \right\}.$$

$$\mathcal{L}_{\text{diff}}(x_0; \phi) := \sum_{i=1}^L \mathbb{E}_{p(x_i|x_0)} \left[ \frac{1}{2 \sigma^2(i)^2} \cdot \underbrace{\|\mu_\phi(x_i, i) - \mu(x_i, x_0, i)\|_2^2}_{\text{SE}(\mu, \mu_\phi, i)} \right]$$

This leads to the final DDPM loss:

$$\mathcal{L}_{\text{DDPM}}(\phi) := \sum_{i=1}^L \frac{1}{2 \sigma^2(i)} \cdot \mathbb{E}_{x_0 \sim p} \mathbb{E}_{p(x_i|x_0)} [\text{SE}(\mu, \mu_\phi, i)]$$

practical choices of predictions and loss

$\xi$ -prediction: (an equivalent reparameterization)

DDPM forward:  $x_i \sim p(x_i | x_0)$  at noise level  $i$  is

$$x_i = \bar{\alpha}_i \cdot x_0 + \sqrt{1 - \bar{\alpha}_i^2} \cdot \xi, \quad x_0 \sim p_{\text{data}}, \quad \xi \sim \mathcal{N}(0, 1)$$

$$\mu(x_i, x, i) = \frac{\bar{\alpha}_{i-1} \cdot \beta_i^2}{1 - \bar{\alpha}_i^2} \quad \times \quad + \quad \frac{(1 - \bar{\alpha}_{i-1}^2) \cdot \bar{\alpha}_i}{1 - \bar{\alpha}_i^2} x_i$$

$$\mu(x_i, x_0, i) = \frac{\bar{\alpha}_{i-1} \cdot \beta_i^2}{1 - \bar{\alpha}_i^2} \cdot \left( \frac{x_i}{\bar{\alpha}_i} - \frac{\sqrt{1 - \bar{\alpha}_i^2}}{\bar{\alpha}_i} \cdot \xi \right) + \frac{(1 - \bar{\alpha}_{i-1}^2) \cdot \bar{\alpha}_i}{1 - \bar{\alpha}_i^2} \cdot x_i$$

$$\text{Note: } \alpha_i = \sqrt{1 - \beta_i^2}, \quad \bar{\alpha}_i = \prod_{k=1}^i \sqrt{1 - \beta_k^2} = \prod_{k=1}^i \alpha_k$$

$$= \left( \frac{1 - \alpha_i^2}{(1 - \bar{\alpha}_i^2) \cdot \bar{\alpha}_i} + \frac{\alpha_i^2 - \bar{\alpha}_i^2}{\bar{\alpha}_i \cdot (1 - \bar{\alpha}_i^2)} \right) x_i - \frac{\bar{\alpha}_{i-1} (1 - \alpha_i^2) \cdot \sqrt{1 - \bar{\alpha}_i^2}}{(1 - \bar{\alpha}_i^2) \cdot \bar{\alpha}_i} \xi$$

$$= \frac{1 - \bar{\alpha}_i^2}{\bar{\alpha}_i \cdot (1 - \bar{\alpha}_i^2)} \cdot x_i - \frac{1}{\bar{\alpha}_i} \cdot \frac{1 - \alpha_i^2}{\sqrt{1 - \bar{\alpha}_i^2}} \cdot \xi$$

$$= \frac{1}{\bar{\alpha}_i} \left( x_i - \frac{1 - \alpha_i^2}{\sqrt{1 - \bar{\alpha}_i^2}} \xi \right)$$

Idea: use  $\mu(x_i, x_0, i) \leftrightarrow \text{model } \mu_\phi(x_i, x_0, i)$  but

$$\xi \leftrightarrow \xi_\phi \leftrightarrow$$

$$\mu_\phi(x_i, i) = \frac{1}{\bar{\alpha}_i} \left( x_i - \frac{1 - \alpha_i^2}{\sqrt{1 - \bar{\alpha}_i^2}} \cdot \xi_\phi(x_i, i) \right)$$

$$\mu(x_i, x_0, i) - \mu_\phi(x_i, i) = -\frac{1-\bar{\alpha}_i^2}{\bar{\alpha}_i(\sqrt{1-\bar{\alpha}_i^2})} \cdot (\xi - \xi_\phi(x_i, i))$$

model now is a "noise detective".

So,

$$\|\mu_\phi(x_i, i) - \mu(x_i, x_0, i)\|_2^2 \propto \|\xi_\phi(x_i, i) - \xi\|_2^2.$$

In practice, the widely used DDPM training loss:

$$\mathcal{L}_{\text{simple}}(\phi) := \mathbb{E}_i \mathbb{E}_{x_0 \sim p_{\text{data}}}( \mathbb{E}_{\xi \sim \mathcal{N}(0,1)} [\|\xi_\phi(x_i, i) - \xi\|^2] )$$

$$x_i = \bar{\alpha}_i x_0 + \sqrt{1-\bar{\alpha}_i^2} \cdot \xi \quad \text{with } x_0 \sim p_{\text{data}}$$

*x*-prediction:

$$\mu(x_i, x, i) = \frac{\bar{\alpha}_{i-1} \cdot \beta_i^2}{1-\bar{\alpha}_i^2} x + \frac{(1-\bar{\alpha}_{i-1}^2) \cdot \bar{\alpha}_i}{1-\bar{\alpha}_i^2} x_i \quad (2.2.4)$$

$x_\phi(x_i, i)$  is trained to predict a clean sample from a given noisy input  $x_i \sim p_i(x_i)$

$$\|\mu_\phi(x_i, i) - \mu(x_i, x_0, i)\|_2^2 \propto \|x_\phi(x_i, i) - x_0\|_2^2$$

$$\Rightarrow \mathbb{E}_i \mathbb{E}_{x_0 \sim p_{\text{data}}} \mathbb{E}_{\xi \sim \mathcal{N}(0,1)} [w_i \|x_\phi(x_i, i) - x_0\|_2^2].$$

[see, Tianlong Li and Kaiming He, 2025, JiT]

x-prediction and  $\epsilon$ -prediction connected via :

$$x_i = \bar{\alpha}_i \cdot x_{\phi}(x_i, i) + \sqrt{1 - \bar{\alpha}_i^2} \epsilon_{\phi}(x_i, i).$$

### DDPM's ELBO

$$p_{\phi}(x_0, x_{1:L}) := \prod_{t=0}^{L-1} p_{\phi}(x_t | x_{t+1}) \cdot p_{\text{prior}}(x_L)$$

$$p_{\phi}(x_0) := \int p_{\phi}(x_0, x_{1:L}) dx_{1:L}$$

#### Theorem 2.2.3: DDPM's ELBO

$$\begin{aligned} -\log p_{\phi}(x_0) &\leq -\mathcal{L}_{\text{ELBO}}(x_0; \phi) \\ &:= \mathcal{L}_{\text{prior}}(x_0) + \mathcal{L}_{\text{recon.}}(x_0; \phi) + \mathcal{L}_{\text{diffusion}}(x_0; \phi) \end{aligned} \quad (2.2.13)$$

Here, each component of losses are defined as:

$$\begin{aligned} \mathcal{L}_{\text{prior}}(x_0) &:= \mathcal{D}_{\text{KL}}(p(x_L | x_0) \| p_{\text{prior}}(x_L)) \\ \mathcal{L}_{\text{recon.}}(x_0; \phi) &:= \mathbb{E}_{p(x_1 | x_0)} [-\log p_{\phi}(x_0 | x_1)] \\ \mathcal{L}_{\text{diffusion}}(x_0; \phi) &= \sum_{i=1}^L \mathbb{E}_{p(x_i | x_0)} \left[ \mathcal{D}_{\text{KL}}(p(x_{i-1} | x_i, x_0) \| p_{\phi}(x_{i-1} | x_i)) \right]. \end{aligned}$$

- $\mathcal{L}_{\text{prior}}$  negligible by choosing  $\{\beta_i\}$  such that

$$p(x_1 | x_0) \approx p_{\text{prior}}$$

- $\mathcal{L}_{\text{recon.}}$  Monte Carlo estimate

- $\mathcal{L}_{\text{diff}}$  we used in diffusion model



**Step 1: Apply Jensen's Inequality.** The marginal log-likelihood is given by:

$$\log p_\phi(\mathbf{x}) = \log \int p_\phi(\mathbf{x}, \mathbf{x}_{0:L}) d\mathbf{x}_0 \cdots d\mathbf{x}_L,$$

with the joint distribution:

$$p_\phi(\mathbf{x}, \mathbf{x}_{0:L}) = p_{\text{prior}}(\mathbf{x}_L) \prod_{i=1}^L p_\phi(\mathbf{x}_{i-1}|\mathbf{x}_i) \cdot p_\phi(\mathbf{x}|\mathbf{x}_0).$$

We introduce the variational distribution  $p(\mathbf{x}_{0:L}|\mathbf{x})$  and rewrite:

$$\log p_\phi(\mathbf{x}) = \log \int p(\mathbf{x}_{0:L}|\mathbf{x}) \frac{p_\phi(\mathbf{x}, \mathbf{x}_{0:L})}{p(\mathbf{x}_{0:L}|\mathbf{x})} d\mathbf{x}_0 \cdots d\mathbf{x}_L.$$

Applying Jensen's inequality ( $\log \mathbb{E}[Z] \geq \mathbb{E}[\log Z]$ ), we obtain the ELBO:

$$\log p_\phi(\mathbf{x}) \geq \mathbb{E}_{p(\mathbf{x}_{0:L}|\mathbf{x})} \left[ \log \frac{p_\phi(\mathbf{x}, \mathbf{x}_{0:L})}{p(\mathbf{x}_{0:L}|\mathbf{x})} \right] =: \mathcal{L}_{\text{ELBO}},$$

and thus,

$$-\log p_\phi(\mathbf{x}) \leq -\mathcal{L}_{\text{ELBO}}.$$

**Step 2: Expand the ELBO.** Assume the variational distribution factorizes as:

$$p(\mathbf{x}_{0:L}|\mathbf{x}) = p(\mathbf{x}_L|\mathbf{x}) \prod_{i=1}^L p(\mathbf{x}_{i-1}|\mathbf{x}_i, \mathbf{x}).$$

Substituting the joint and variational distributions into the ELBO:

$$\begin{aligned} \mathcal{L}_{\text{ELBO}} &= \mathbb{E}_{p(\mathbf{x}_{0:L}|\mathbf{x})} \left[ \log p_{\text{prior}}(\mathbf{x}_L) + \sum_{i=1}^L \log p_\phi(\mathbf{x}_{i-1}|\mathbf{x}_i) + \log p_\phi(\mathbf{x}|\mathbf{x}_0) \right. \\ &\quad \left. - \log p(\mathbf{x}_L|\mathbf{x}) - \sum_{i=1}^L \log p(\mathbf{x}_{i-1}|\mathbf{x}_i, \mathbf{x}) \right]. \end{aligned}$$

We now compute the negative ELBO by grouping terms according to their dependencies and applying marginalization:

$$\begin{aligned} -\mathcal{L}_{\text{ELBO}} &= \mathbb{E}_{p(\mathbf{x}_0|\mathbf{x})} [-\log p_\phi(\mathbf{x}|\mathbf{x}_0)] + \mathbb{E}_{p(\mathbf{x}_L|\mathbf{x})} \left[ \log \frac{p(\mathbf{x}_L|\mathbf{x})}{p_{\text{prior}}(\mathbf{x}_L)} \right] \\ &\quad + \sum_{i=1}^L \mathbb{E}_{p(\mathbf{x}_i|\mathbf{x})} \left[ \mathbb{E}_{p(\mathbf{x}_{i-1}|\mathbf{x}_i, \mathbf{x})} \left[ \log \frac{p(\mathbf{x}_{i-1}|\mathbf{x}_i, \mathbf{x})}{p_\phi(\mathbf{x}_{i-1}|\mathbf{x}_i)} \right] \right]. \end{aligned}$$

To justify the last term, we use the factorization:

$$p(\mathbf{x}_i, \mathbf{x}_{i-1}|\mathbf{x}) = p(\mathbf{x}_i|\mathbf{x}) \cdot p(\mathbf{x}_{i-1}|\mathbf{x}_i, \mathbf{x}),$$

which leads to:

$$\begin{aligned} &\mathbb{E}_{p(\mathbf{x}_{0:L}|\mathbf{x})} \left[ \log \frac{p(\mathbf{x}_{i-1}|\mathbf{x}_i, \mathbf{x})}{p_\phi(\mathbf{x}_{i-1}|\mathbf{x}_i)} \right] \\ &= \int p(\mathbf{x}_i|\mathbf{x}) \left[ \int p(\mathbf{x}_{i-1}|\mathbf{x}_i, \mathbf{x}) \log \frac{p(\mathbf{x}_{i-1}|\mathbf{x}_i, \mathbf{x})}{p_\phi(\mathbf{x}_{i-1}|\mathbf{x}_i)} d\mathbf{x}_{i-1} \right] d\mathbf{x}_i \\ &= \mathbb{E}_{p(\mathbf{x}_i|\mathbf{x})} \left[ \mathbb{E}_{p(\mathbf{x}_{i-1}|\mathbf{x}_i, \mathbf{x})} \left[ \log \frac{p(\mathbf{x}_{i-1}|\mathbf{x}_i, \mathbf{x})}{p_\phi(\mathbf{x}_{i-1}|\mathbf{x}_i)} \right] \right]. \end{aligned}$$

We may refer to the three terms in  $-\mathcal{L}_{\text{ELBO}}$  as:

$$\mathcal{L}_{\text{recon.}}, \quad \mathcal{L}_{\text{prior}}, \quad \mathcal{L}_{\text{diffusion}},$$

corresponding respectively to the reconstruction loss, the prior KL, and the stepwise diffusion KL. This completes the derivation.  $\blacksquare$

## DDPM Sampling:

- $\xi_{\phi^x}(x_i, i)$   $i=1, 2, \dots, L$  ← after training
- $x_L \sim p_{\text{prior}} = \mathcal{N}(0, I)$  : sampling procedure

$$x_L \xrightarrow{p_{\phi^x}(x_{L-1}|x_L)} x_{L-1} \xrightarrow{p_{\phi^x}(x_{L-2}|x_{L-1})} x_{L-2} \dots \xrightarrow{p_{\phi^x}(x_0|x_1)} x_0$$

$$x_{i-1} \leftarrow \underbrace{\frac{1}{\bar{\alpha}_i} \left( x_i - \frac{1-\bar{\alpha}_i^2}{\sqrt{1-\bar{\alpha}_i^2}} \xi_{\phi^x}(x_i, i) \right)}_{\mu_{\phi^x}(x_i, i)} + \sigma(i) \cdot \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, I)$$

- Interpretation of DDPM's sampling:

$$x_i \leftarrow \text{interpolation between } \underline{x_i} \text{ and } \underline{x_{\phi^x}(x_i, i)} + \sigma(i) \cdot \epsilon_i$$

$$\Leftrightarrow x_{i-1} = A \cdot x_i + B \cdot x_{\phi^x}(x_i, i) + \sigma(i) \cdot \epsilon_i$$

**proof:** By  $x_i = \bar{\alpha}_i \cdot x_{\phi^x}(x_i, i) + \sqrt{1-\bar{\alpha}_i^2} \epsilon_{\phi^x}(x_i, i)$

$$\Rightarrow x_{\phi^x}(x_i, i) = \frac{x_i - \sqrt{1-\bar{\alpha}_i^2} \cdot \epsilon_{\phi^x}(x_i, i)}{\bar{\alpha}_i}$$

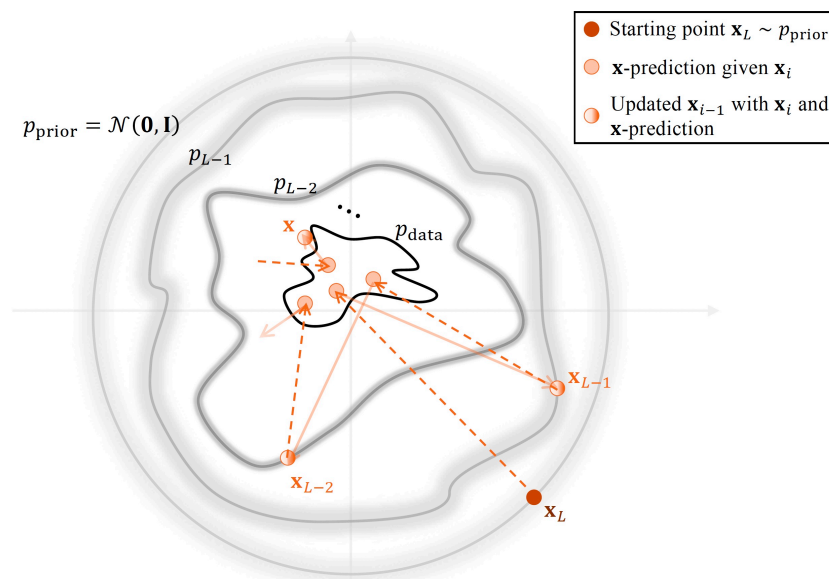
Rearranging to isolate  $\epsilon_{\phi^x}$ :

$$\epsilon_{\phi^x}(x_i, i) = \frac{x_i - \bar{\alpha}_i \cdot x_{\phi^x}(x_i, i)}{\sqrt{1-\bar{\alpha}_i^2}}$$

$$\Rightarrow \mu_{\phi^x}(x_i, i) = \underbrace{\left( \frac{1}{\bar{\alpha}_i} - \frac{1-\bar{\alpha}_i^2}{\bar{\alpha}_i(1-\bar{\alpha}_i^2)} \right)}_A \cdot x_i + \underbrace{\left( \frac{(1-\bar{\alpha}_i^2) \cdot \bar{\alpha}_i}{\bar{\alpha}_i(1-\bar{\alpha}_i^2)} \right)}_B \cdot x_{\phi^x}(x_i, i)$$

DDPM can be viewed as an iterative denoising process:

- (1): Estimating clean data  $x_{\phi^*}(x_i, i)$  from current noisy input  $x_i$ .
- (2): Sampling less noisy latent  $x_{i-1}$  via the update rule using this clean estimate.



### Remarks of sampling via DDPM:

- (1): Blurry predictions at high noise levels

$$x^*(x_i, i) = \mathbb{E}[x_0 | x_i], \quad x_i \sim p_i$$

the model predict the expected clean data given  $x_i$

- (2): Early steps set the global structure  
Later steps add fine detail

- (3): slow sampling:

$L = 1000$  steps in practice

- (4) In practice,  $p_{\phi}(x_{i-1} | x_i)$  modeled as Gaussian

} small  $\beta_i \rightarrow$  true reverse distribution closes to Gaussian  
{ large  $\beta_i \rightarrow$  induce strong non-Gaussian.

<https://lilianweng.github.io/posts/2018-08-12-vae/>

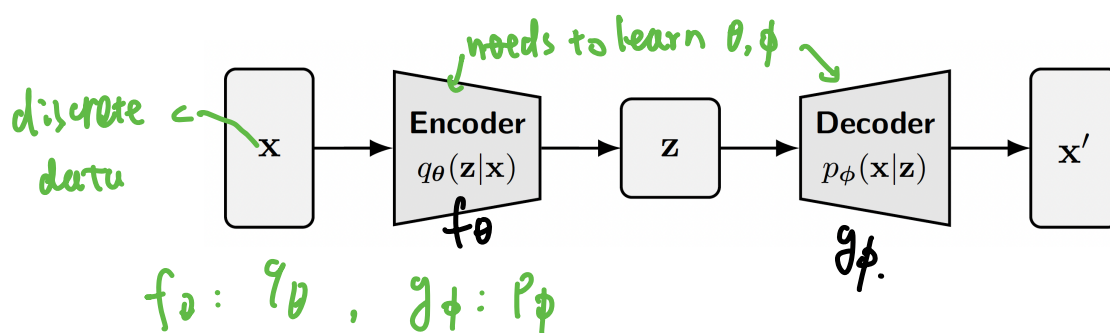
① AE - Autoencoder

② VAE - Variational Autoencoder

③ DDPM - Denoising Diffusion Probabilistic Model

④ Discrete - VAE

⑤ D3PMs



$x \sim \text{Lat}(p)$   $p = [p_1, p_2, \dots, p_k]$ ,  $\sum_{i=1}^k p_i = 1$

$x \sim \text{Categorical distribution}$  PMF:  $f_x(x) = \begin{cases} p_1 & x=1 \\ p_2 & x=2 \\ \vdots & \vdots \\ p_k & x=k \end{cases}$

$f_x(x) = \sum_{i=1}^k [x=i]$ ,  $[\text{statement}] = \begin{cases} 1 & \text{if true} \\ 0 & \text{otherwise} \end{cases}$

$x \rightarrow z$ : discrete latent variable  $x \in \{1, 2, \dots, k\}$

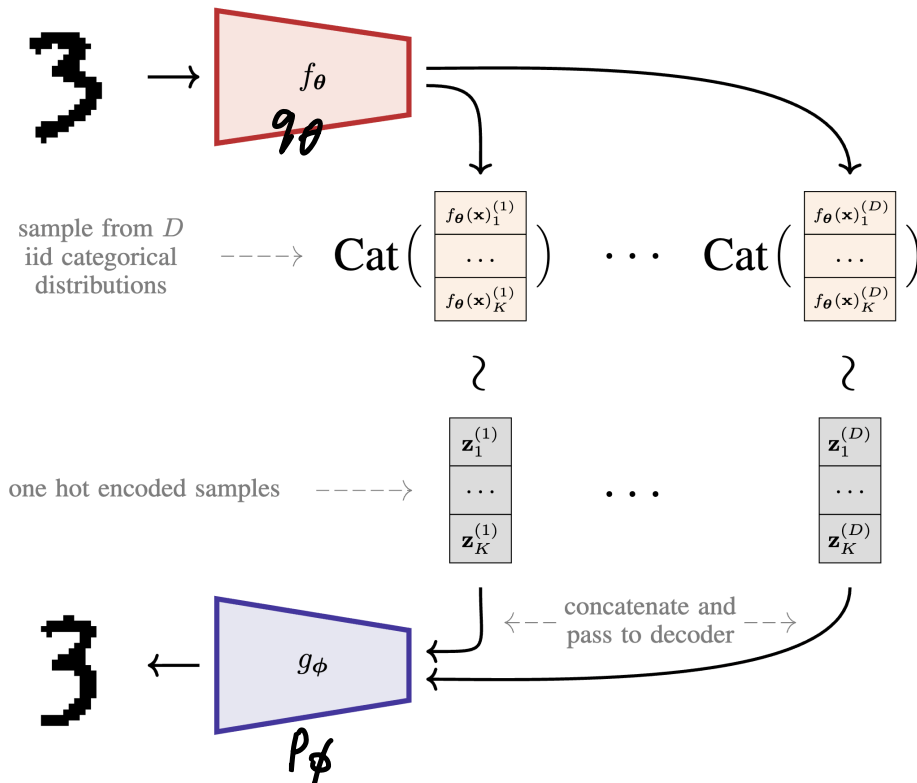
$z \in \{0, 1\}^{d \times k}$ .  $f_{\theta}: \mathbb{R}^d \rightarrow \mathbb{R}^{k \times d}$   $d$ : latent variables.

$z_i \sim \mathcal{D}^{(i)}(f_{\theta}(x)^{(i)})$  for  $i=1, 2, \dots, d$

$$\sum_{k=1}^k z_k^{(i)} = 1, \quad \forall i=1, 2, \dots, d.$$

$$K^{-1} = [\frac{1}{K}, \dots, \frac{1}{K}]$$

natural prior distribution  $p(z) \sim \text{Uniform}(K) = \text{Lat}(K^{-1})$ .



$$\mathcal{L}_{ELBO} = \mathbb{E}_{z \sim q_\theta(z|x)} [\log p_\phi(x|z)] - D_{KL}(q_\theta(z|x) \parallel p(z))$$

$$\begin{cases} \nabla_\theta := \nabla_\theta \mathcal{L}_{ELBO} \\ \nabla_\phi := \nabla_\phi \mathcal{L}_{ELBO} \end{cases} \quad x \in \{0,1\}^D: \text{binary image matrix}$$

$$\nabla_\phi = \nabla_\phi \mathbb{E}_{z \sim q_\theta(z|x)} [\log p_\phi(x|z)]$$

$$= \mathbb{E}_{z \sim q_\theta(z|x)} [\nabla_\phi \log p_\phi(x|z)]$$

$$\nabla_\phi \log p_\phi(x|z) = \nabla_\phi \log \prod_{i=1}^D p_\phi(x^{(i)}|z)$$

$$= \sum_{i=1}^D \nabla_\phi \log p_\phi(x^{(i)}|z)$$

$$x \in \{0,1\}^D. \text{ PMF: } f_x(x) = p^x \cdot (1-p)^{4-x}$$

$$\log p_{\phi}(x^{(i)}|z) = x^{(i)} \cdot \log p_{\phi}(x^{(i)}=1|z) + (1-x^{(i)}) \log p_{\phi}(x^{(i)}=0|z)$$

$$\nabla_{\phi} \log p_{\phi}(x|z) = - \sum_{i=1}^n \nabla_{\phi} \cdot \text{CE}(p^{(i)}, x^{(i)}).$$

[see. Alan Jeffares and Liyuan Liu, 2025]

① AE - Autoencoder

② VAE - Variational Autoencoder

③ DDPM - Denoising Diffusion Probabilistic Model

④ Discrete - VAE

⑤ D3PMs

### ○ The Idea of D3PM

Let  $x_t, x_{t-1}$  be scalar discrete random variables with  $K$  categories [ $K$ ] :=  $\{1, 2, \dots, K\}$ . Forward transition probabilities can be represented by matrices:

$$[Q_t]_{ij} = p(x_t = j | x_{t-1} = i).$$

$x_t$  and  $x_{t-1}$  can be represented as row vectors, then we have

$$p(x_t | x_{t-1}) = \text{Cat}(x_t; x_{t-1} Q_t)$$

- $\text{Cat}(x; \cdot)$  is a categorical distribution
- $Q_t$  is applied to each pixel of an image or each token in a sequence
- $Q_t$  is applied independently:  $x_t = [x_t^{(1)}, x_t^{(2)}, \dots, x_t^{(T)}]$ , then for each  $t$ :  $p(x_t | x_{t-1}) = \prod_{i=1}^T p(x_t^{(i)} | x_{t-1}^{(i)})$

$t$ -step marginal and posterior at time  $t-1$  is

$$p(x_t | x_0) = \text{Cat}(x_t; x_0 \bar{Q}_t), \quad \bar{Q}_t = Q_1 Q_2 \dots Q_t$$

$$p(x_{t+1} | x_t, x_0) = \frac{p(x_t | x_{t-1}, x_0) \cdot p(x_{t+1} | x_t)}{p(x_t | x_0)} = \text{Cat}(x_{t+1}; \frac{x_t Q_{t+1}^T \odot x_0 \bar{Q}_{t+1}}{x_0 \bar{Q}_t \cdot x_t^T}),$$



where note:  $q(x_t | x_{t-1}, x_0) = q(x_t | x_{t-1})$  due to the Markov property.

We want to have  $p_\phi(x_t | x_{t-1}) \simeq p(x_t | x_{t-1})$ , we can efficiently train  $p_\phi$  when  $p(x_t | x_{t-1})$ :

- ① efficient sampling of  $x_t$  from  $q(x_t | x_0)$  for any  $t$
- ② a tractable expression for forward posterior  $p(x_t | x_t, x_0)$

• Forward process  $p(x_{1:L} | x_0) = \prod_{t=1}^L p(x_t | x_{t-1})$ ,  $x_0 \sim p_{\text{data}}(x)$

$x_{1:L} = x_1, x_2, \dots, x_L$  increasingly noisy

• Reverse Markov process  $p_\phi(x_{0:L}) = p(x_L) \cdot \prod_{t=1}^L p_\phi(x_{t-1} | x_t)$

The whole loss:

$$\begin{aligned}
 L_{vb} &= \mathbb{E}_{x_0 \sim p_{\text{data}}(x)} \left[ \underbrace{D_{KL}[p(x_L | x_0) \| p(x_L)]}_{L_T} \right] + \\
 &\sum_{t=2}^T \mathbb{E}_{x_t \sim p(x_t | x_0)} \left[ \underbrace{D_{KL}[p(x_{t-1} | x_t, x_0) \| p_\phi(x_{t-1} | x_t)]}_{L_{t-1}} \right] + \\
 &\quad - \mathbb{E}_{x_1 \sim p(x_1 | x_0)} \left[ \log p_\phi(x_0 | x_1) \right] \\
 &= \mathbb{E}_{x_0 \sim p_{\text{data}}(x)} \left[ \underbrace{\sum_{t=0}^T L_t}_{L_0} \right]
 \end{aligned}$$

### ○ The choice of $Q_t$

- rows of  $Q_t$  must be sum to 1 = probability mass
- rows of  $\bar{Q}_t = Q_1 Q_2 \dots Q_t$  must converge to a known

stationary distribution:

given any distribution  $\pi$ ,  $P$  is said to be stationary if

$$\pi = \pi \cdot P. \text{ This means } \pi = \pi \cdot P^n.$$

(1) Uniform:  $Q_t = (1-\beta_t) \cdot I + \beta_t \cdot I \cdot I^T / \kappa$  with  $\beta_t \in [0,1]$

- $Q_t$  is doubly stochastic

- $\pi = \pi \cdot Q_t$ : the stationary distribution is uniform

(2). Absorbing: by adding [MASK] token

$x_t \xrightarrow{Q} \begin{cases} x_t : \text{stay the same} \\ \text{[MASK]} \text{ with some probability } \beta_t \end{cases}$

$$[Q_t]_{ij} = \begin{cases} 1 & \text{if } i=j=m \\ 1-\beta_t & \text{if } i=j \neq m \\ \beta_t & \text{if } j=m, i \neq m \end{cases} = (1-\beta_t) \cdot I + \beta_t \cdot I \cdot e_m^T$$

(3) Discretized Gaussian:  $\frac{4(i-j)^2}{(k-1)^2 \beta_t}$

$$[Q_t]_{ij} = \begin{cases} \frac{\exp(-\frac{4(i-j)^2}{(k-1)^2 \beta_t})}{\sum_{n=-k+1}^{k-1} \exp(-\frac{4n^2}{(k-1)^2 \beta_t})} & \text{if } i \neq j \\ 1 - \sum_{l=0, l \neq i}^{k-1} [Q_t]_{il} & \text{if } i=j \end{cases}$$

(4) Using word-embedding distance to have locality

$$[G_t]_{ij} = \begin{cases} =1 & \text{if } w_i \text{ is a } k\text{-nearest neighbor of } w_j \\ =0 & \text{otherwise} \end{cases} : k\text{-nearest neighbor}$$

Define  $A = \frac{(G_t + G_t^T)}{2k}$

$$[R]_{ij} = \begin{cases} -\sum_{l \neq i} A_{il} & \text{if } i=j \\ A_{ij} & \text{otherwise} \end{cases}$$

$$Q_t = \exp(\alpha_t R) = \sum_{n=0}^{\infty} \frac{\alpha_t^n}{n!} R^n$$